

Laporan Praktikum Jaringan Komputer IF-04-02

MODUL 10 IP

Tujuan : Mahasiswa dapat menginvestigasi cara kerja protokol IP menggunakan Wireshark

Menangkap paket dari eksekusi traceroute >tracert gaia.cs.umass.edu

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>tracert gaia.cs.umass.edu

Tracing route to gaia.cs.umass.edu [128.119.245.12]
over a maximum of 30 hops:

  1  16 ms    3 ms     3 ms  10.218.0.253
  2  19 ms    14 ms    10 ms  10.217.15.254
  3  25 ms     7 ms     4 ms  36.67.147.33
  4  34 ms    26 ms     *    180.240.190.101
  5  *         *         *    Request timed out.
  6  41 ms    24 ms    24 ms  180.240.204.8
  7  60 ms    26 ms    26 ms  be4126.ccr81.sin01.atlas.cogentco.com [154.18.3.249]
  8  197 ms   193 ms   191 ms  be9040.ccr41.lax04.atlas.cogentco.com [154.54.87.145]
  9  191 ms   192 ms   200 ms  be3271.ccr41.lax01.atlas.cogentco.com [154.54.42.101]
 10  204 ms   204 ms   203 ms  be2931.ccr31.phx01.atlas.cogentco.com [154.54.44.85]
 11  208 ms   209 ms   211 ms  be5471.ccr21.elp02.atlas.cogentco.com [154.54.166.57]
 12  222 ms   222 ms   221 ms  be3821.ccr31.dfw01.atlas.cogentco.com [154.54.165.25]
 13  236 ms   240 ms   235 ms  be5383.ccr31.oma02.atlas.cogentco.com [154.54.7.234]
 14  242 ms   243 ms   243 ms  be5214.ccr41.ord01.atlas.cogentco.com [154.54.165.134]
 15  248 ms   247 ms   248 ms  port-channel2717.ccr91.cle04.atlas.cogentco.com [154.54.6.222]
 16  250 ms   251 ms   252 ms  be8022.ccr21.alb02.atlas.cogentco.com [154.54.169.186]
 17  267 ms   259 ms   261 ms  be8077.ccr71.orh02.atlas.cogentco.com [154.54.169.213]
 18  262 ms   262 ms   264 ms  be2191.ccr71.orh01.atlas.cogentco.com [154.54.160.70]
 19  262 ms   264 ms   261 ms  38.32.33.83
 20  275 ms   262 ms   262 ms  69.16.0.9
 21  276 ms   261 ms   266 ms  69.16.3.0
 22  280 ms   265 ms   265 ms  core2-rt-et-4-3-0.gw.umass.edu [192.80.83.105]
 23  275 ms   272 ms   271 ms  n5-rt-1-1-et-10-0-0.gw.umass.edu [128.119.0.10]
 24  268 ms   269 ms   268 ms  128.119.77.235
 25  *         *         *    Request timed out.
 26  266 ms   267 ms   267 ms  cics-rt-xe-0-0-0.gw.umass.edu [128.119.3.32]
 27  *         *         *    Request timed out.
 28  266 ms   270 ms   265 ms  gaia.cs.umass.edu [128.119.245.12]

Trace complete.

C:\Users\fairuuz>
```

```
*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 10.218.10.7 && ip.dst == 128.119.245.12

No. Time Source Destination Protocol Length Info
---
2132 79.6305360.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133 79.6466705.. 10.218.10.7 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2134 79.6494364.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135 79.6593904.. 10.218.10.7 128.119.245.12 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2136 79.6553062.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137 79.6584837.. 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2342 85.6231090.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2343 85.6422256.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2344 85.6432761.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2345 85.6580224.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2346 85.6584230.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2347 85.6684418.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2568 91.4050117.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=68/17408, ttl=3 (no response found!)
2569 91.4301371.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2570 91.4334573.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=69/17664, ttl=3 (no response found!)
2571 91.4406517.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2572 91.5399319.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=70/17920, ttl=3 (no response found!)
2573 91.5447210.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)

Frame 2132: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: RuijieNetwork cf:9d:26 (4
Internet Protocol Version 4, Src: 10.218.10.7, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 92
Identification: 0x99ee (39406)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: ICMP (1)
Header Checksum: 0x954e [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.218.10.7
Destination Address: 128.119.245.12
[Stream index: 14]
Internet Control Message Protocol

Packets: 6203 - Displayed: 155 (2.5%) - Dropped: 0 (0.0%)
Profile: Default
```

Wi-Fi

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ip.src == 10.218.10.7 && ip.dst == 128.119.245.12

No.	Time	Source	Destination	Protocol	Length	Info
2132	79.6305360...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133	79.6466705...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2134	79.6494364...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135	79.6523984...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2136	79.6553062...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137	79.6584837...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2342	85.6231090...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2343	85.6422256...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2344	85.6432761...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2345	85.6580224...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2346	85.6584230...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2347	85.6684418...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2568	91.4050117...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=68/17408, ttl=3 (no response found!)
2569	91.4301171...	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2570	91.4334573...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=69/17664, ttl=3 (no response found!)
2571	91.4406517...	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2572	91.5399319...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=70/17920, ttl=3 (no response found!)
2573	91.5447210...	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

Frame 2133: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface 0

Ethernet II, Src: RuijieNetwork cf:9d:26 (48:81:d4:cf:9d:26), Dst: LiteonTechno_af:cf:7d (2)

Internet Protocol Version 4, Src: 10.218.0.253, Dst: 10.218.10.7

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 120

Identification: 0xba35 (47669)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x9ed8 [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.218.0.253

Destination Address: 10.218.10.7

[Stream index: 15]

Internet Control Message Protocol

wireshark_Wi-FiQZRW03.pcapng

Packets: 6203 - Displayed: 155 (2.5%) - Dropped: 0 (0.0%)

Profile: Default

IPv4 Dasar

TTL 1 :

Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
2132	79.6305360...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133	79.6466705...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2134	79.6494364...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135	79.6523984...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2136	79.6553062...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137	79.6584837...	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2155	80.0900239...	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2201	81.6041055...	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2258	83.1113684...	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2342	85.6231090...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2343	85.6422256...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2344	85.6432761...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2345	85.6580224...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2346	85.6584230...	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2347	85.6684418...	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2351	85.6957205...	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2405	87.1884189...	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2451	88.6899068...	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)

Frame 2132: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: RuijieNetwork_cf:9d:26 (4)

Internet Protocol Version 4, Src: 10.218.10.7, Dst: 128.119.245.12

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 92

Identification: 0x99ee (39406)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 1

Protocol: ICMP (1)

Header Checksum: 0x954e [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.218.10.7

Destination Address: 128.119.245.12

[Stream index: 14]

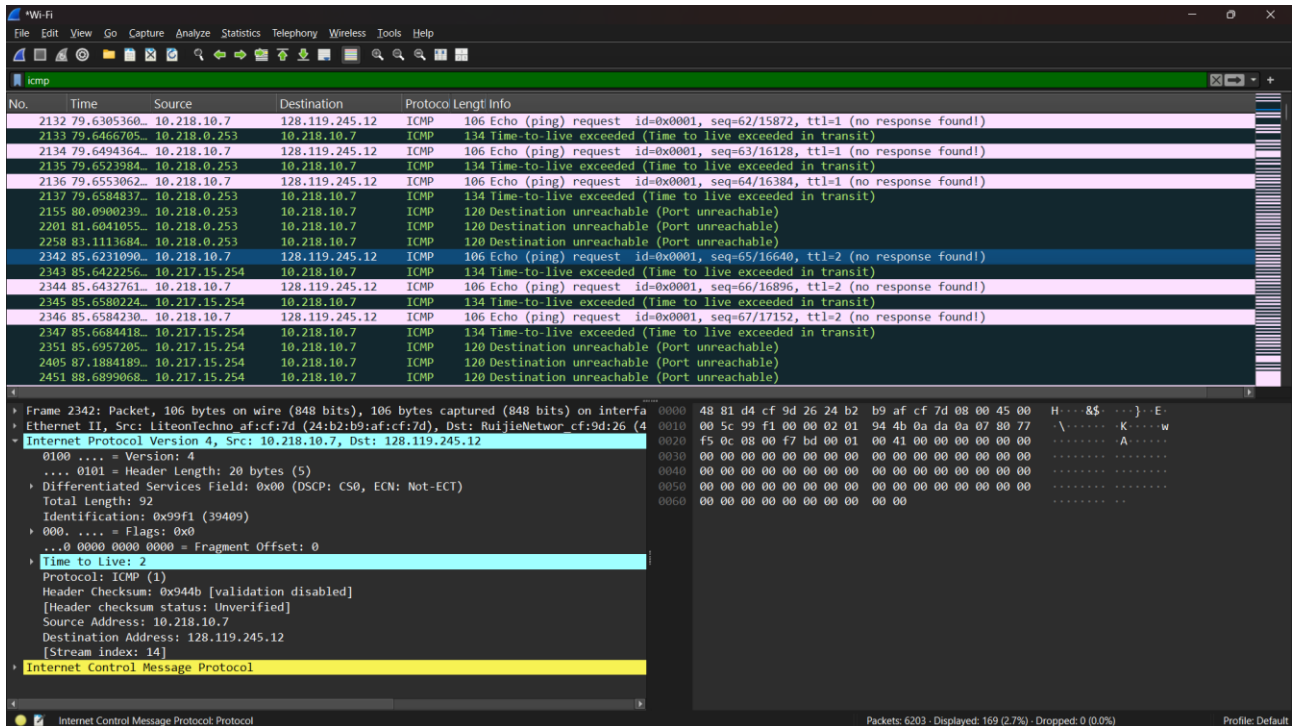
Internet Control Message Protocol

Internet Control Message Protocol: Protocol

Packets: 6203 - Displayed: 169 (2.7%) - Dropped: 0 (0.0%)

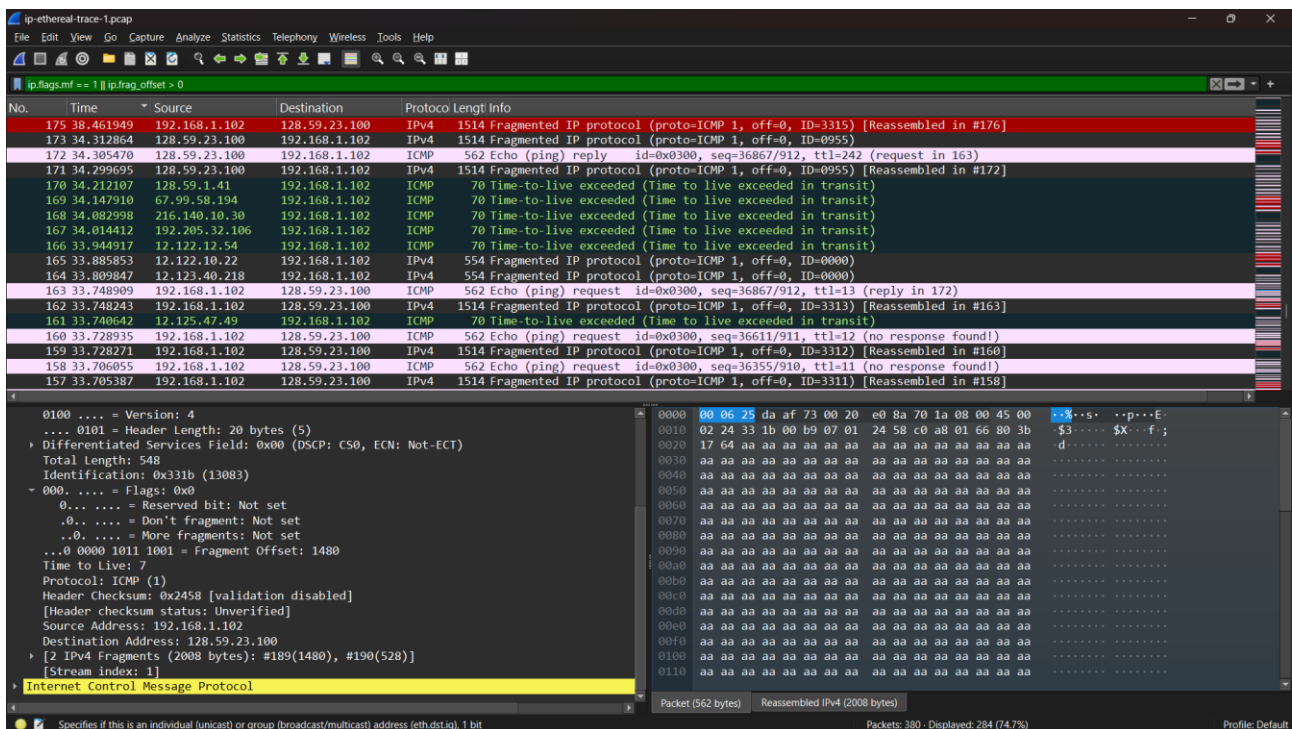
Profile: Default

TTL 2 :



Pada hasil capture Wireshark, terlihat bahwa komputer mengirim ICMP Echo Request dengan nilai TTL = 1 ke tujuan 128.119.245.12. Router pertama kemudian mengurangi nilai TTL menjadi 0 dan mengirimkan balasan ICMP “Time-To-Live Exceeded”. Hal ini menunjukkan bahwa traceroute berhasil mengidentifikasi hop pertama dalam jaringan.

Fregmentasi



Pada hasil analisis ditemukan fragmentasi IP. Hal ini ditunjukkan oleh adanya paket dengan nilai Fragment Offset lebih dari 0, yaitu sebesar 1480. Selain itu, nilai More Fragments (MF) pada paket

tersebut adalah 0, yang menunjukkan bahwa paket tersebut merupakan fragment terakhir. Fragmentasi terjadi karena datagram yang dikirim melebihi ukuran Maximum Transmission Unit (MTU), sehingga dipecah menjadi beberapa bagian. Hal ini diperkuat dengan adanya beberapa paket yang memiliki nilai Identification yang sama.

IPv6

Berdasarkan hasil analisis menggunakan Wireshark, ditemukan paket DNS yang berisi permintaan tipe A (IPv4), seperti pada domain youtube.com dan doubleclick.net. Namun, tidak ditemukan paket DNS dengan tipe AAAA (IPv6). Hal ini menunjukkan bahwa pada saat proses capture dilakukan, sistem hanya melakukan resolusi alamat IPv4 dan tidak melakukan permintaan alamat IPv6. Kemungkinan hal ini disebabkan oleh konfigurasi jaringan atau DNS server yang tidak mengutamakan penggunaan IPv6.

